PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Emmanuel John
Program/Batch	B084-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (theHarvester based Footprinting Attacks) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report presents the practical activities completed during Week 2 of my Cybersecurity and Ethical Hacking internship at Networkwalks. The activities covered two key areas: **Footprinting & Reconnaissance with theHarvester (W2-PM1)** and **Zenmap-Based Network Scanning (W2-PM5)**.

The first activity focused on gathering publicly available information about the `networkwalks.com` domain using various Kali Linux reconnaissance tools, including theHarvester. The second activity involved using Zenmap, a graphical interface for Nmap, to scan my own local network and identify active hosts.

Both activities provided practical exposure to the initial stages of a cybersecurity assessment, from collecting information about a target domain to discovering devices on a network. This report documents the commands used, the results observed, supporting screenshots, and the significance of the findings from a cybersecurity perspective.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
nslookup	Resolve the domain name to its IP address using DNS.
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
theHarvester tool	An open-source OSINT (Open-Source Intelligence) tool used in cybersecurity to gather information about a target domain from publicly available sources

4. Activities Performed

4.1 Footprinting & Reconnaissance

theHarvester

I used **theHarvester** to perform reconnaissance against the **microsoft.com** domain. The command `theHarvester -d microsoft.com -l 50 -b all` was executed to gather publicly available information from multiple search engines and data sources.

The tool attempted to collect information such as subdomains, hostnames, IP addresses, and email addresses associated with the target domain.

During the scan, several data sources were accessed, including Baidu, DuckDuckGo, CertSpotter, crt.sh, Common Crawl, HackerTarget, GitLab, LeakIX, and RapidDNS.

However, the scan encountered limitations because several services required API keys that were not configured. Some other sources returned connection timeouts, invalid responses, or no response.

The Hudson Rock search completed successfully but returned **0 hosts, 0 IP addresses, 0 email addresses, and 0 stealer records**. Other sources, including Robtex, also failed to return responses.

Overall, theHarvester demonstrated how multiple public intelligence sources can be used to gather information about a domain. However, the results were limited by missing API keys and connectivity issues, so the scan did not produce a complete set of reconnaissance findings.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap**, the graphical user interface for Nmap, to perform network discovery on my local network. The objective of this activity was to identify active devices connected to the network, determine their IP addresses, and understand how network discovery can be used to map devices within a network.

I began by identifying the local network configuration and determining the IP address range to be scanned. I then used Zenmap to perform a **Ping Scan** against the selected network subnet. This scan was intended to identify which IP addresses were associated with active hosts.

The scan was executed using Nmap 7.993 and covered a total of **256 IP addresses**. The results identified one active host on the network:

- **IP Address:** 192.168.56.1
- **Host Status:** Up (Active)
- **Total IP Addresses Scanned:** 256
- **Total Live Hosts Discovered:** 1
- **Scan Duration:** 12.36 seconds

The scan confirmed that the device with IP address **192.168.56.1** was reachable and responding to the network discovery scan. The remaining 255 IP addresses were not identified as active hosts during this scan.

The results demonstrated how Nmap can systematically examine a network range and identify active devices. This information is useful during network reconnaissance because it helps establish which hosts are accessible within a network and provides an initial understanding of the network environment.

After completing the scan, I reviewed the results in Zenmap. The practical also involved using the **Topology** tab to visualize discovered hosts and understand their relationship within the network. The topology view can help provide a graphical representation of the discovered network devices.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Limited Host Discovery	The scan identified only 1 active host (192.168.56.1) out of 256 IP addresses scanned.	Other devices may not have been identified, resulting in an incomplete view of the network.	● **Low**
2	Active Host Identified	Host 192.168.56.1 was reported as up and responding to the scan.	The device represents an active network endpoint that should be included in network inventory and monitoring.	● Informational
3	Missing MAC Address Information	The scan output did not provide a MAC address for the discovered host.	The physical network device cannot be identified using the available scan results alone.	● Informational
4	Limited Host Information	The scan did not provide device names, operating systems, or running services.	Further network assessment may be required to understand the devices and services present on the network.	● **Low**

Risk level key: ● Critical ● Informational ● Low

The risks identified above are observations from the network discovery and scanning exercise, not confirmed vulnerabilities.

The practical exercise primarily focused on identifying active hosts within the local network using Zenmap and Nmap. The scan successfully identified one active host, 192.168.56.1, out of 256 IP addresses scanned.

No vulnerability assessment, exploitation, or security validation was performed as part of this activity.

Therefore, identifying an active IP address or having limited information about a network device does not automatically indicate a security weakness. Further authorized network assessment would be required to identify and confirm any actual vulnerabilities.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Maintain an updated network inventory**
Organizations should maintain an accurate record of devices connected to their networks, including their IP addresses, MAC addresses and device names.

2. **Perform regular network discovery**
Internal network scans should be conducted periodically to identify active hosts and maintain visibility of connected devices.

3. **Investigate unidentified devices**
Any unfamiliar or unexpected devices discovered during network scanning should be investigated to verify whether they are authorized to access the network.

4. **Monitor network activity**
Organizations should regularly monitor network traffic and device activity to identify unusual or unexpected connections.

5. **Document network topology**
Network diagrams should be created and updated regularly to provide a clear understanding of connected devices and network infrastructure.

6. **Verify network configurations**
IP addressing, subnet configurations and network settings should be reviewed to ensure devices are properly configured.

7. **Improve device identification**
Organizations should collect relevant device information, including hostnames and MAC addresses, to support effective network management.

8. **Conduct regular security assessments**
Periodic authorized security assessments should be performed to identify potential weaknesses and improve the overall security of the network.

9. **Perform scanning with proper authorization**
Network discovery and scanning activities should only be conducted on systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed two practical activities: Footprinting & Reconnaissance and Zenmap-Based Network Scanning.

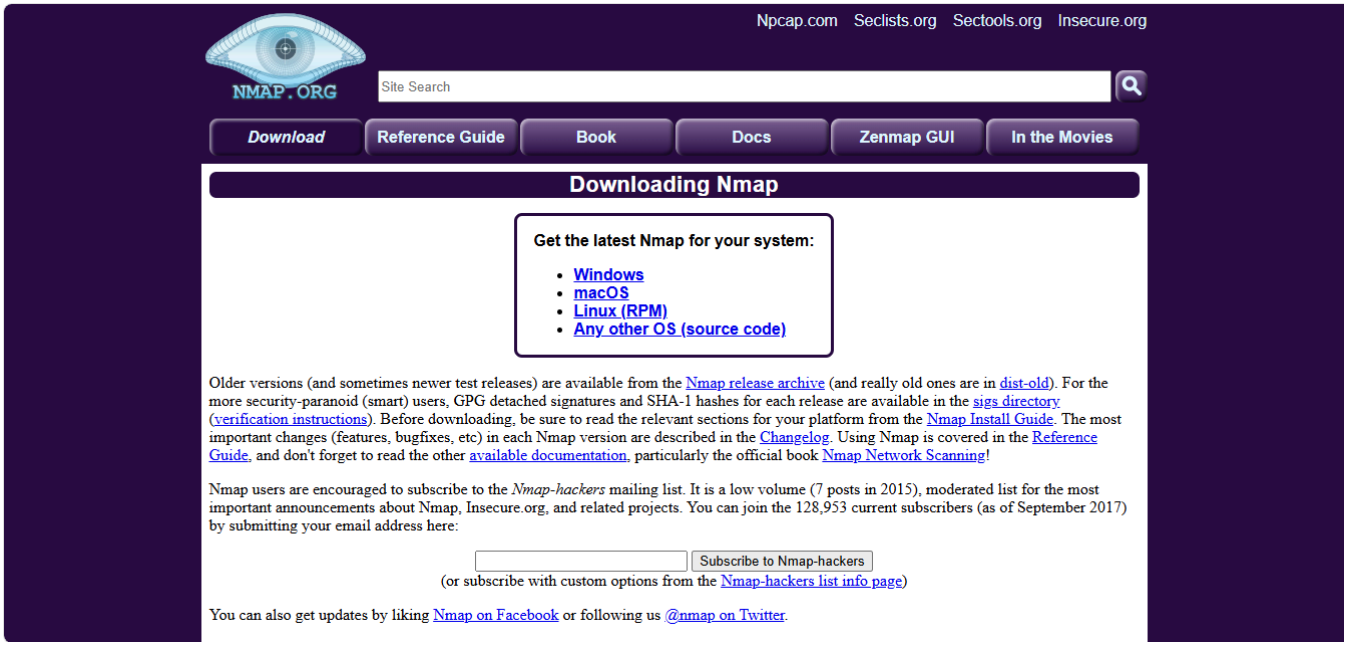
The footprinting and reconnaissance activity introduced me to different Kali Linux tools used to gather information about a target domain. I explored domain registration details, web technologies, DNS information, HTTP response headers and Web Application Firewall identification. I also used theHarvester to gather publicly available information from multiple sources and learned how missing API keys and connection issues can affect reconnaissance results.

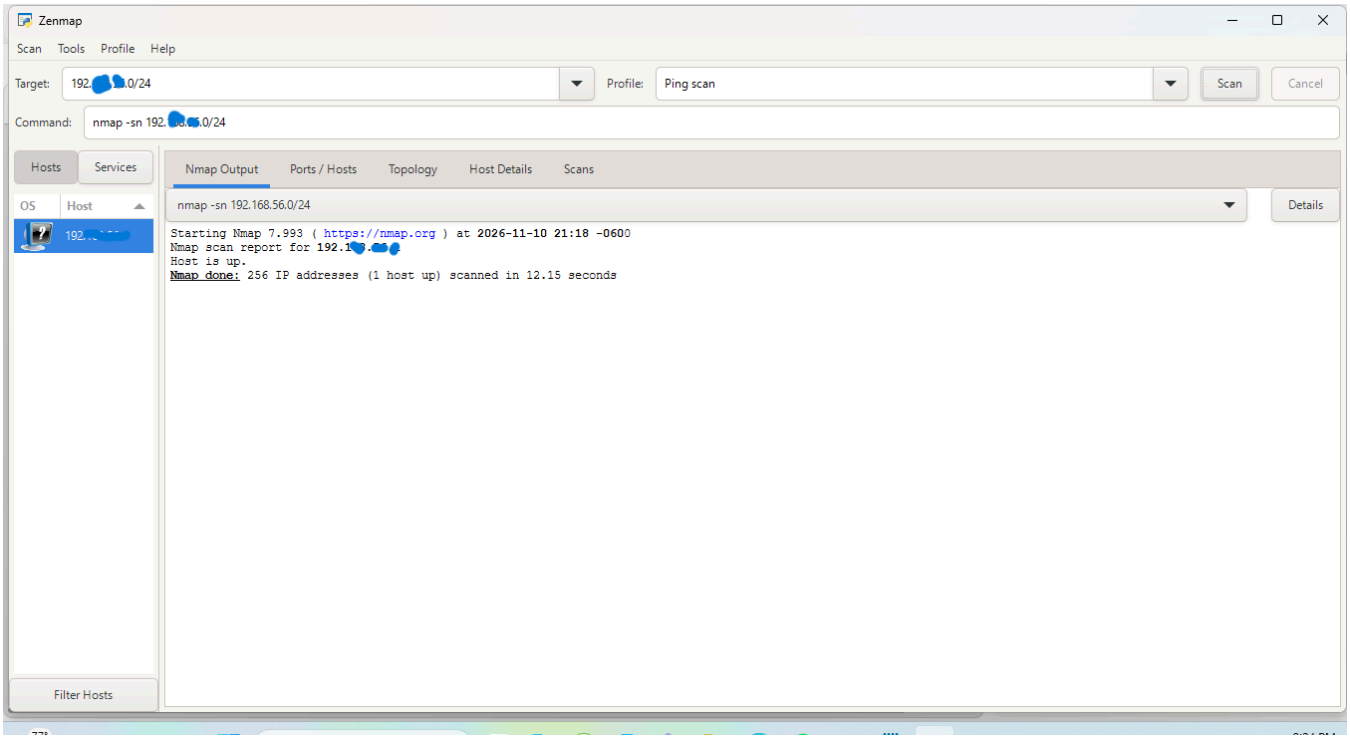
For the network scanning activity, I used Zenmap and Nmap to perform a scan of 256 IP addresses. The scan identified one active host, **192.168.56.1**, demonstrating how network scanning can be used to discover live hosts within a network.

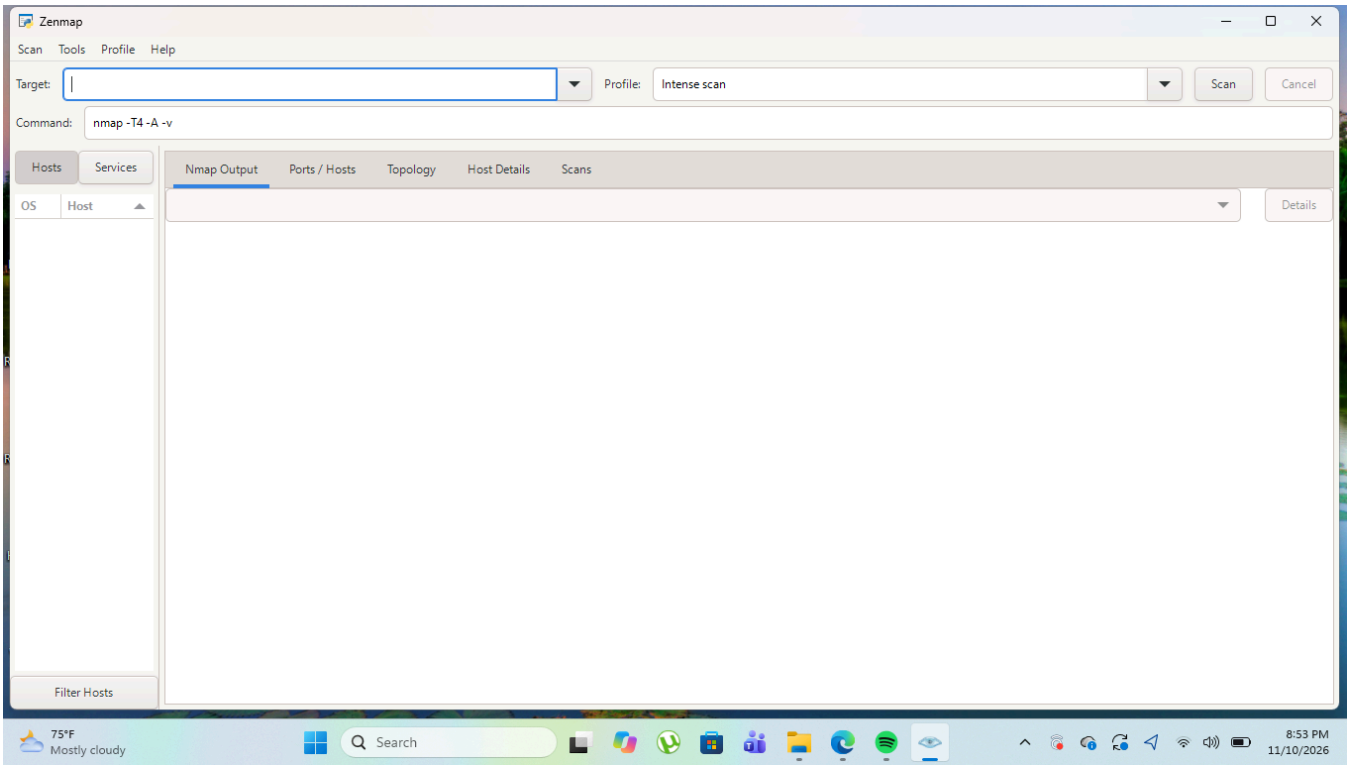
Overall, these activities improved my understanding of the initial stages of cybersecurity assessments. I gained practical exposure to reconnaissance tools, network discovery and the interpretation of technical results. I also learned the importance of documenting findings accurately, identifying potential risks and ensuring that all reconnaissance and scanning activities are performed within an authorized environment.

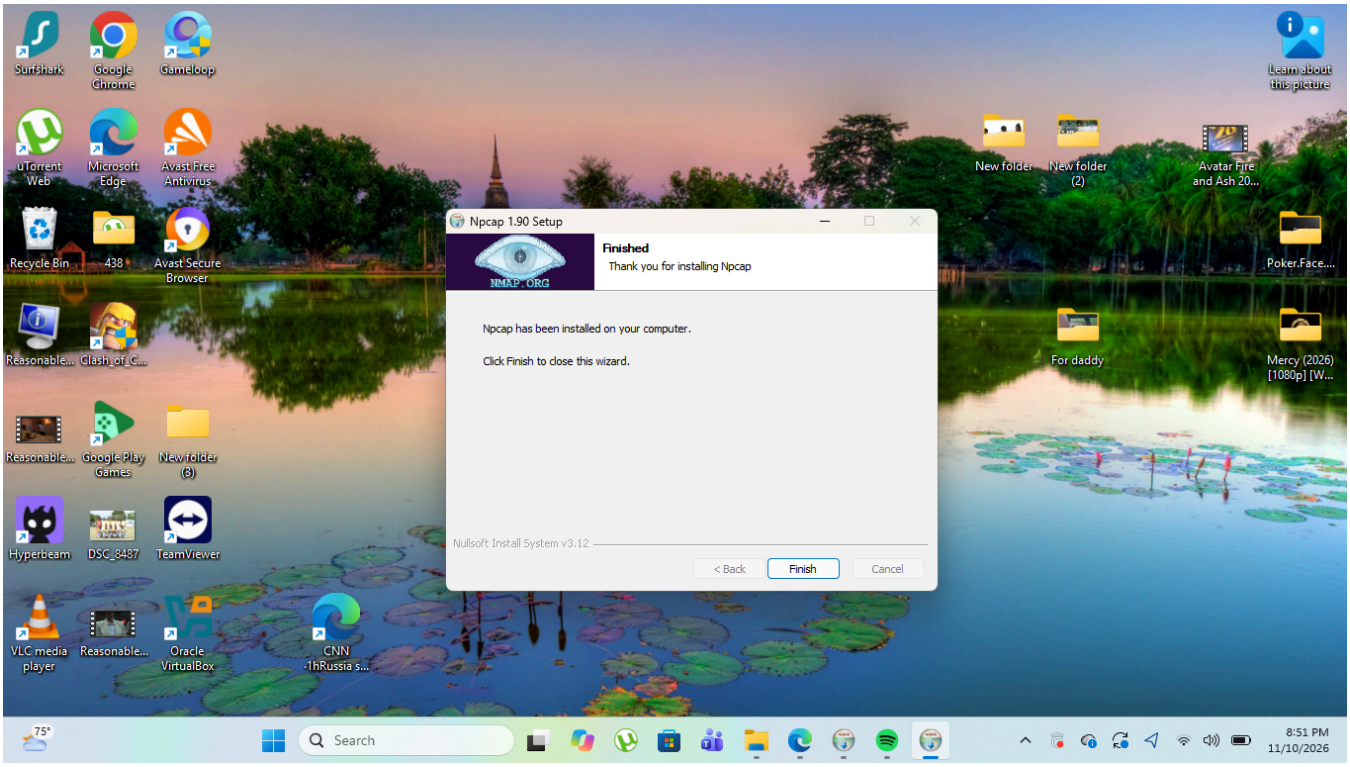
These exercises provided a valuable foundation for developing my practical cybersecurity and ethical hacking skills.

8. Evidences Collected

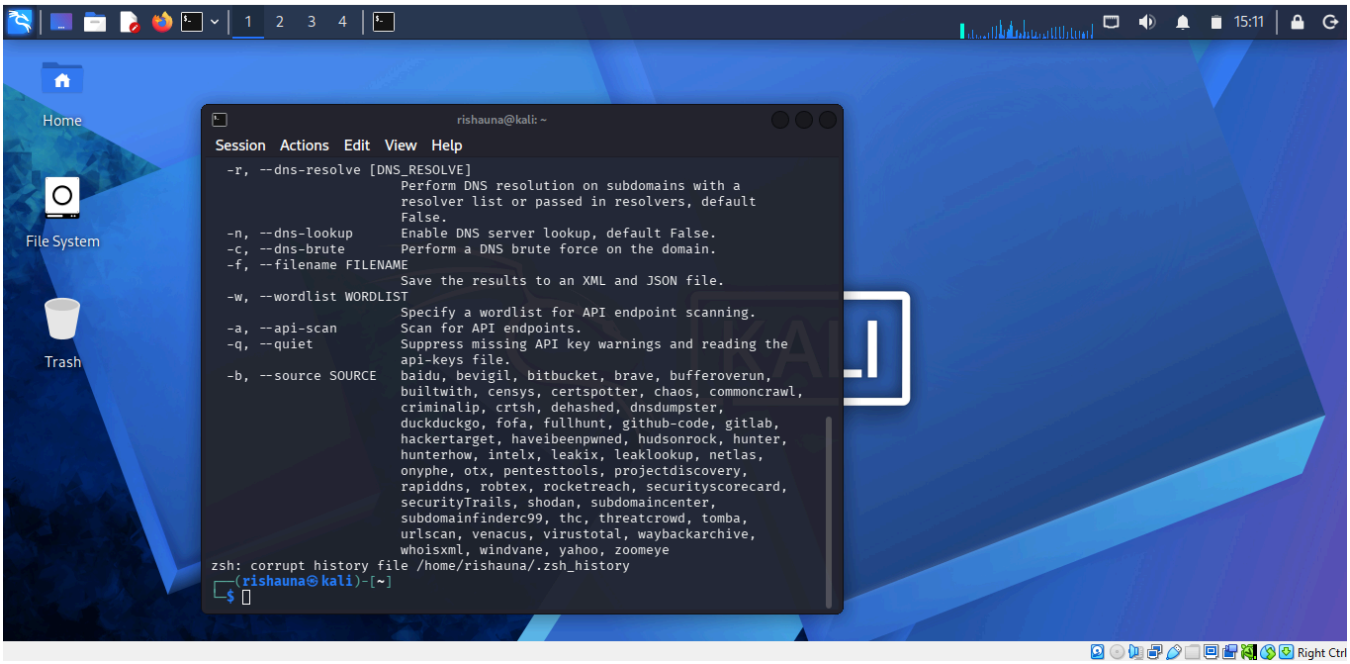




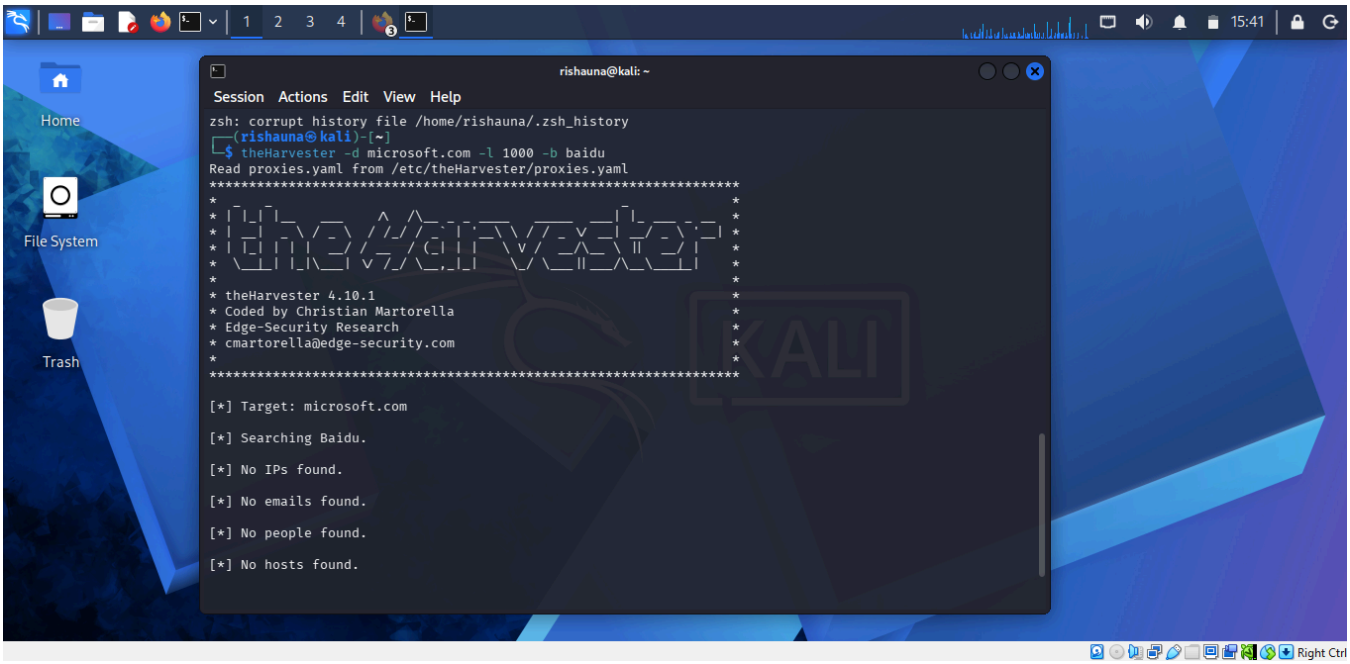

Zenmap
Scan Tools Profile Help
Target:
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v
Hosts
Services
OS
Host
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Details
Filter Hosts
75°F
Mostly cloudy
Search
8:53 PM
11/10/2026

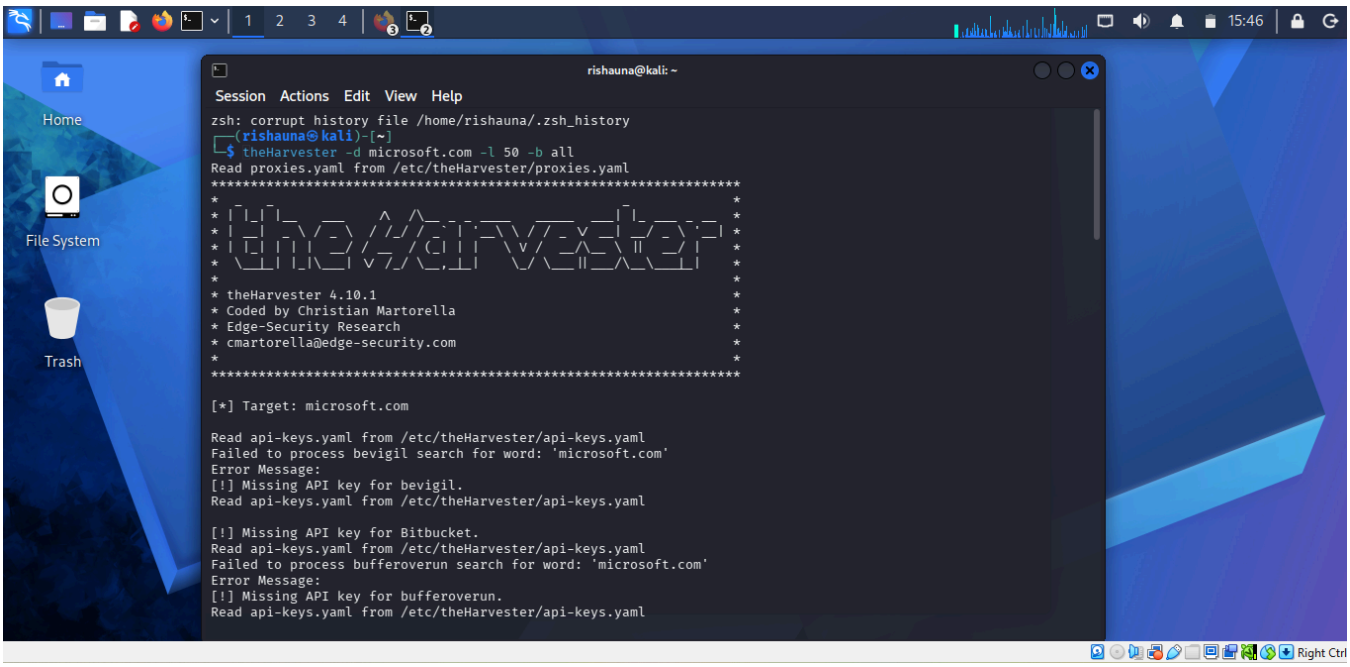

Surfshark
Google Chrome
Gameloop
Learn about this picture
uTorrent Web
Microsoft Edge
Avast Free Antivirus
New folder
New folder (2)
Avatar Fire and Ash 20...
Recycle Bin
438
Avast Secure Browser
Poker.Face....
Reasonable...
Clash_of_C...
For daddy
Mercy (2026) [1080p] [W...
Reasonable...
Google Play Games
New folder (3)
Hyperbeam
DSC_8487
TeamViewer
VLC media player
Reasonable...
Oracle VirtualBox
CNN -1hRussia s...
Npcap 1.90 Setup
NMAP.ORG
Finished
Thank you for installing Npcap
Npcap has been installed on your computer.
Click Finish to close this wizard.
Nullsoft Install System v3.12
< Back
Finish
Cancel
Search
8:51 PM
11/10/2026

-End-


1 2 3 4
15:11
Home
File System
Trash
rishauna@kali: ~
Session Actions Edit View Help
-r, --dns-resolve [DNS_RESOLVE]
Perform DNS resolution on subdomains with a
resolver list or passed in resolvers, default
False.
-n, --dns-lookup Enable DNS server lookup, default False.
-c, --dns-brute Perform a DNS brute force on the domain.
-f, --filename FILENAME
Save the results to an XML and JSON file.
-w, --wordlist WORDLIST
Specify a wordlist for API endpoint scanning.
-a, --api-scan Scan for API endpoints.
-q, --quiet Suppress missing API key warnings and reading the
api-keys file.
-b, --source SOURCE baidu, bevigil, bitbucket, brave, bufferoverun,
builtwith, censys, certspotter, chaos, commoncrawl,
criminalip, crtsh, dehashed, dnsdumpster,
duckduckgo, fofa, fullhunt, github-code, gitlab,
hackertarget, haveibeenpwned, hudsonrock, hunter,
hunterhow, intelx, leakix, leaklookup, netlas,
onyphe, otx, pentesttools, projectdiscovery,
rapiddns, robtex, rocketreach, securityscorecard,
securityTrails, shodan, subdomaincenter,
subdomainfinderc99, thc, threatcrowd, tomba,
urlscan, venacus, virustotal, waybackarchive,
whoisxml, windvane, yahoo, zoomeye
zsh: corrupt history file /home/rishauna/.zsh_history
(rishauna㉿kali)-[~]
$
Right Ctrl


theHarvester 4.10.1
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cmartorella@edge-security.com





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📌 Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub